

Canada based PhD candidate in Computational and Cognitive Neuroscience at the Hebrew University of Jerusalem, focusing on visual attention in naturalistic conditions. Former software engineer at Facebook (Meta), with a BSc. in Computer Science at Tel Aviv University. I am proficient in eye-tracking and EEG analysis, statistics, data science and software engineering. I have a proven ability to lead complex projects from planning through execution, both as a team leader and a collaborative team member.

Education

- 2022-present **Hebrew University (PhD Candidate): [ELSC Direct PhD Track](#) in Computational Neuroscience**
Advisor: Prof. Leon Deouell, MD PhD
- Course GPA: 95.8
- 2022-2025 **Hebrew University (MSc): Computational Neuroscience (Direct PhD Track)**
Advisor: Prof. Leon Deouell, MD PhD
- Final GPA: 90.1
 - Developed and published an open-source Python package for benchmarking eye-tracking data parsers.
 - Experienced in co-registration and analysis of large-scale multi-modal physiological and behavioral data.
- 2016-2020 **Tel-Aviv University (BSc): Computer-Science & Psychology (Emphasis on Neuroscience)**
- Final GPA: 86.5
 - BSc. Thesis: *A no-report paradigm for probing human memory using eye tracking*.
Advisors: Omer Sharon, PhD; Prof. Yuval Nir, PhD
 - BSc. Thesis: *Face recognition during free viewing of naturalistic video stimuli*.
Advisor: Prof. Galit Yovel, PhD
 - Studied a semester in Konstanz University, Germany (Mar-Aug 2018).
- 2007-2010 **"Ironi Hey" High School (Haifa)**
- Majored in Math, English, Physics and Chemistry at the highest level.

Publications

- 2026
- Nir, J. (2026). Data Sharing and Citation Impact in Visual Search Research. *PsyArXiv*, 2026-04. [pre-print]
 - Nir, J. & Deouell, L.Y. (2026). Systematic Classification Differences Across Eye-Movement Detection Algorithms. *Behavior Research Methods* 58, 109.
- 2025
- Schmidig, F. J., ..., Nir, J., ... & Nir, Y. (2025). Anticipatory eye gaze as a marker of memory. *Communications Psychology*, 3(1), 122.

Conferences

- 2026
- Nir, J., Auerbach-Asch R.C., & Deouell, L.Y. (2026, May 15-19). *Fixated but Failed to See: Recognition Errors in Visual Search*. Annual Meeting of the Vision Science Society ([VSS 2026](#)), St. Pete Beach, FL, USA. [poster presentation]
- 2025
- Nir, J. & Deouell, L.Y. (2025, Jul. 6-9). *pEYES: An Open-Source Python Toolkit for Eye Tracking Data Analysis and Detector Evaluation*. Association for the Scientific Study of Consciousness Annual Meeting ([ASSC 28](#)), Heraklion, Greece. [poster presentation]
 - Nir, J. & Deouell, L.Y. (2025, Feb. 25-27). *A Comparative Evaluation of Eye Movement Detection Algorithms*. Israeli Society for Cognitive Psychology Annual Conference ([ISCOP 11](#)), Acre, Israel. [poster presentation]

Awards & Scholarships

- 2022-present • ELSC Excellence Scholarship for Direct-Track PhD Students ([info](#)).
- 2024 • Israeli National Road Safety Authority (רלב"ד) research grant for MSc students ([info](#)).
- Israeli Smart Transportation Research Center (ISTRC) scholarship for MSc students ([info](#)).
- 2018 • Erasmus+ scholarship for Israeli students studying in European universities ([info](#)).
- 2015 • Military-Intelligence's Major-General award for Creative Thinking.

Professional Experience

- 2022-2023 **Academic Teaching:** Hebrew University
 - Teaching Assistant: Summer course for graduate students in neuroscience from the Hebrew University & Technical University of Munich (76953; September 2023, HUJI+TUM)
 - Lab Instructor: Introduction to Machine Learning ([67577](#); spring semester 2023, HUJI)
 - Teaching Assistant: Statistical Methods ([51307](#); winter semester 2022, HUJI)
- 2019-2022 **Software Engineer:** Express Wi-Fi (Facebook TLV)
 - Unblocked deployment in multiple regions by integrating our product (multi-service, AWS) with FB's internal SMS platform, while maintaining data separation required for GDPR compliance.
 - Increased ad-impressions by 30% by adjusting existing customer-flows and supporting new ad-types.
 - Team PoC for experimentation processes: design, implementation & review.
- 2017-2019 **Research Assistant & Software Developer:** Yuval Nir Lab (Tel-Aviv University)
 - Implemented industry-standard feature-extraction algorithms for eye-tracking data (python).
 - Conducted clinical trials on humans to collect physiological measures, such as EEG, eye tracking, etc.

Languages and Skills

- **Hebrew** – native; **English** – fluent.
- Programming languages: python, Go, javascript & React.
- Experiment planning, execution and data analysis.
- Fast learning, independent and responsible.
- Good presentational & interpersonal skills.
- Hobbies: Yoga, CrossFit, Hiking, Cooking.

Example Projects

- *pEYES*, a package for eye-tracking algorithm evaluation and analysis [python; [github](#)]
- Implementing SLAM on KITTI dataset using visual odometry, bundle adjustment & gtsam [python; [github](#)]